KENYA MEDICAL TRAINING COLLEGE
FACULTY OF INFORMATION TECHNOLOGY & MEDICAL EDUCATION
DEPARTMENT OF HEALTH RECORDS & INFORMATION

# FINAL QUALIFYING EXAMINATION
# FOR
# DIPLOMA IN HEALTH RECORDS & INFORMATION TECHNOLOGY

## PAPER: HEALTH STATISTICS

| | |
|---|---|
| **DATE: 10th February, 2020** | **TIME: 3 HOURS (9:00AM – 12:00NOON)** |

**INSTRUCTIONS**

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the EXAMINATION NUMBER given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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7. Which of the following is NOT TRUE?

   a) A chi square test measures how expectations compare to actual observed data
   b) Degrees of freedom is just the number of categories
   c) Independent samples in a t-test compares the means for two groups
   d) In probability distribution, the mean is a product of probability for success and the sample size

8. The proportion for whom data is obtained is referred to as

   a) Population
   b) Sample size
   c) Response rate
   d) Random sample

9. Which of the following is not true about quantitative data?

   a) Are numerical
   b) Can arise from counts
   c) Can arise from measurements
   d) May fall in separate classes

10. The count of individuals having a particular quality can be referred to as

   a) Frequency distribution
   b) Array
   c) Frequency
   d) Discrete

11. _____ for value of a variable is the number of individuals with values less than or equal to that value

   a) Frequency
   b) Cumulative frequency
   c) Relative frequency
   d) Relative cumulative frequency

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## SECTION 1

## MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. The following are probability distributions except
   a) Binomial distribution
   b) Frequency distribution
   c) Poisson distribution
   d) Normal distribution
2. Another name for chain referral sampling method is
   a) Snowball
   b) Purposive
   c) Convenience
   d) Quota
3. The size of a sample depends on the following except
   a) The indicators chosen
   b) The baseline value of the indicators
   c) The amount of change you would want to measure accurately
   d) The geographical area
4. Which of the following is not a feature of a standard normal curve
   a) Its mean is 1
   b) Its standard deviation is 1
   c) Its total area is 1
   d) Its symmetrical and bell shaped
5. The following are examples of errors during sampling except
   a) Error of origin
   b) Biased error
   c) Sampling error
   d) Non sampling error
6. Which one of the following is NOT true about primary sources of data?
   a) It gives very reliable data
   b) It is very cheap to conduct
   c) There is a possibility of sampling errors
   d) It has a short-term value



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12. _____ is a diagram where the class intervals are on an axis and rectangles with heights or areas are proportional to the frequencies erected on them

a) Histogram

b) Ogive

c) Cumulative frequency polygon

d) Frequency polygon

13. A positively skewed distribution can also be referred to as

a) Skew downwards

b) Skew to the right

c) Skew to the left

d) Skew upwards

14. The measure of spread of a distribution is called

a) Central tendency

b) Measure of location

c) Interquartile range

d) Dispersion

15. _____ is calculated from the squares of observation

a) Standard deviation

b) Coefficient of variation

c) Variance

d) Mean

16. Which quantity best describes degrees of freedom

a) $n-5$

b) $n-10$

c) $n-1$

d) $n-0$

17. A bar chart can also be referred as

a) Bar graph

b) Bar presentation

c) Bar size

d) Bar diagram



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18. The probability must lie between

a) -1 to 1

b) 0.1 – 1

c) 0.5 – 1

d) 0 – 1

19. The implication rule is probability applies to _____events

a) Independent events

b) Mutually exclusive events

c) Mutually inclusive events

d) Normal events

20. _____distribution is followed by the number of events which happen in a fixed tune interval

a) Normal distribution

b) Binomial distribution

c) Frequency distribution

d) Poisson distribution

21. Gaussian distribution is another name for

a) Binomial distribution

b) Poisson distribution

c) Normal distribution

d) Frequency distribution

22. One of the following is not a type of mean

a) Arithmetic mean

b) Real mean

c) Harmonic mean

d) Geometric mean

23. _____estimates errors caused by sampling

a) Chi square

b) Student t test

c) F test

d) Confidence interval



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24. The hypothesis of no difference in the population is called

a) Alternative hypothesis

b) Alternate hypothesis

c) Null hypothesis

d) True hypothesis

25. The probability of an extreme value of a test statistic occurring if the null hypothesis is true is called

a) Confidence interval

b) Statistical significant

c) P value

d) Significant level

26. _____estimates the numerical relationship between variables

a) Correlation

b) Regression

c) Regression estimate

d) Correlation estimate

27. _____is a method which tells us about the nature of the relationship between two variables

a) Regression

b) Hypothesis

c) Deviation

d) Correlation

28. _____scale individuals are grouped but not necessarily ordered

a) Interval scale

b) Nominal scale

c) Ration scale

d) Ordinal scale

29. Another name for a contingency table is

a) Cross classification table

b) Non classification table

c) Frequency table

d) Classification table



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30. What is the probability of obtaining a HEAD when a coin is tossed?

a) 2

b) 0

c) 0.5

d) 1

31. One of the following is not a type of missing data

a) Missing completely at random

b) Missing partially at random

c) Missing at random

d) Missing not at random

32. When constructing a Z chart, the following need to be considered except

a) Percentage monthly figures

b) Moving annual totals

c) Individual monthly figures

d) Monthly cumulative figures for the year

33. _____sampling relies on arranging the target population according to some ordering scheme and then selecting elements at regular intervals through the ordered list

a) Systematic

b) Multi stage

c) Stratified

d) Quota

34. Data from central bureau of statistics is an example of

a) Primary data

b) Secondary data

c) Original data

d) Tertiary data

35. The following are advantages of a focused group discussion except

a) Responses of each participant are independent

b) Provide an opportunity to involve people in data analysis

c) Quick cheap and relatively easy to assemble

d) People are able to build on one another's responses and come up with good ideas



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36. Which one of the following is not categorical data?

a) Gender

b) Age

c) Choice of test

d) Marital status

37. The number of patients occupying a hospital bed at any given time is called

a) Bed occupancy

b) Bed state

c) Bed turnover

d) Turn over per bed

38. Another name for quartile deviation is

a) Interquartile range

b) Semi interquartile range

c) Quartile range

d) Uni quartile range

39. The following are advantages of range except

a) Sample to calculate

b) Sample to understand

c) Representative in case of extreme values

d) Used as a measure of dispersion in statistical quality control

40. The general equation for any straight line on a graph is

a) $y = b + ax$

b) $x = b + ax$

c) $y = a + bx$

d) $x = a + bx$

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## SECTION 2

## SHORT ANSWERS QUESTIONS (40 MARKS)

1. State five types of events in probability **(5 marks)**
2. List five properties of normal distribution **(5 marks)**
3. Calculate the number of classes required for the following data

   9, 5, 6, 12, 18, 25, 32, 46, 8, 7, 10, 12, 16, 17, 21, 24, 26, 29, 32, 35, 38 **(5 marks)**
4. Briefly describe the two types of numerical data **(5 marks)**
5. Outline 5 important points to consider while constructing a histogram **(5 marks)**
6. List any 5 measures of dispersion **(5 marks)**
7. Briefly explain any 2 methods of data collection **(5 marks)**
8. List five non random sampling methods **(5 marks)**

## SECTION 3

## LONG ANSWER QUESTIONS (20 MARKS)

1. a) In a study to determine the adoption of EMR systems in Maragua Sub county hospital, 150 health professionals were studied. Determine the sample size if the proportion of the characteristic being measured was 30%. Use the fishers et al formula at 95% confidence level. **(10 marks)**

   b) Pona hospital had a bed complement of 150 and an average length of stay of 7 days. The hospital had a daily occupancy of 119 with bed turn over of 57 patients. At discharge each patient paid 250 shillings per day as fees.

   Calculate

   i. Percentage occupancy **(3 marks)**

   ii. Total discharges and death **(3 marks)**

   iii. Turn over interval **(2 marks)**

   iv. Total fees collected in the year **(2 marks)**



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